

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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A proof of the central limit theorem

- Let $X_1, X_2, \dots, X_N$ be i.i.d. random variables that form a random sample of size 'N'. Assume that we have drawn this sample from a population that has a mean μ and variance σ^2 .
- Let $\bar{X}_N$ be the sample mean: $\bar{X}_N = \frac{X_1 + X_2 + \dots + X_N}{N}$
- Let $\bar{Z}_N$ be the standardized sample mean: $\bar{Z}_N = \frac{\bar{X}_N - \mu}{\sigma/\sqrt{N}}$
- The Central Limit Theorem states that as N tends to infinity, $\bar{Z}_N$ converges in distribution to $N(0, 1)$, i.e. the CDF of $\bar{Z}_N$ becomes identical to the CDF of $N(0, 1)$.
- To prove this statement, we use the property of the Moment Generating Function (MGF) that if the MGFs of X and Y are identical, then so are their CDFs.

Moment Generating Function

- The k-th moment of a random variable $\mathbf{X}$ is the expected value of $\mathbf{X}$ raised to the k-th power:

$$\mu_k(\mathbf{X}) = \mathbb{E}(\mathbf{X}^k) = \sum_i x_i^k P(\mathbf{X} = x_i)$$
- The k-th moment of $\mathbf{X}$ around some value $\mathbf{c}$ is known as the k-th central moment of $\mathbf{X}$:

$$\mu_k(\mathbf{X}) = \mathbb{E}((\mathbf{X} - \mathbf{c})^k) = \sum_i (x_i - \mathbf{c})^k P(\mathbf{X} = x_i)$$
- The k-th standardized moment of $\mathbf{X}$ is the k-th central moment of $\mathbf{X}$ divided by k-th power of the standard deviation of $\mathbf{X}$: $\frac{\mu_k(\mathbf{X})}{\sigma^k} = \frac{\mathbb{E}((\mathbf{X}-\mathbf{c})^k)}{\sigma^k}$
- For the record, the first 5 moments of $\mathbf{X}$ have specific values or meanings attached to them:
 - The zeroth's raw and central moments of $\mathbf{X}$ are $\mathbb{E}(\mathbf{X}^0)$ and $\mathbb{E}[(\mathbf{X}-\mathbf{c})^0]$ respectively. Both equate to 1.
 - The 1st raw moment of $\mathbf{X}$ is $\mathbb{E}(\mathbf{X})$. It's the mean of $\mathbf{X}$.
 - The second central moment of $\mathbf{X}$ around its mean is $\mathbb{E}[\mathbf{X}-\mathbb{E}(\mathbf{X})]^2$. It's the variance of $\mathbf{X}$.
 - The third and fourth standardized moments of $\mathbf{X}$ are $\mathbb{E}[\mathbf{X}-\mathbb{E}(\mathbf{X})]^3/\sigma^3$, and $\mathbb{E}[\mathbf{X}-\mathbb{E}(\mathbf{X})]^4/\sigma^4$. They are the skewness and kurtosis of $\mathbf{X}$ respectively.
- Let's now define a new random variable $t\mathbf{X}$ where t is a real number. Here's the Taylor series expansion of e to the power $t\mathbf{X}$ evaluated at $t = 0$:

$$e^{t\mathbf{X}} = \sum_{k=0}^{\infty} \frac{(t\mathbf{X})^k}{k!} = 1 + \frac{t}{1!}\mathbf{X} + \frac{t^2}{2!}\mathbf{X}^2 \dots$$

- If we apply the *Expectation operator* on both sides of the above equation we get from the linearity of the operator:

$$\mathbb{E}(e^{t\mathbf{X}}) = \sum_{k=0}^{\infty} \mathbb{E}\left(\frac{(t\mathbf{X})^k}{k!}\right) = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}(\mathbf{X}^k)$$

- If we now write the general form of an *Exponential Generating Function* (EGF):

$$\sum_{k=0}^N \frac{a_k}{n!} x^k = a_0 + a_1 x + \frac{a_2}{2!} x^2 + \dots$$

- We see that $\mathbb{E}(\mathbf{X}^k)$ are the coefficients a_k in the EGF. Thus we have the *Moment Generating Function* (MGF):

$$M_{\mathbf{X}}(t) = \mathbb{E}(e^{t\mathbf{X}}) = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}(\mathbf{X}^k) = 1 + t\mathbb{E}(\mathbf{X}) + \frac{t^2}{2!} \mathbb{E}(\mathbf{X}^2) + \dots$$

- Now, it is easy to realize that the k-th derivative of the EGF, when evaluated at $x = 0$ gives us the k-th coefficient of the underlying sequence. So,

$$M_{\mathbf{X}}^0(t=0) = 1, M_{\mathbf{X}}^1(t=0) = \mathbb{E}(\mathbf{X}), \dots, M_{\mathbf{X}}^k(t=0) = \mathbb{E}(\mathbf{X}^k)$$
- If two random variables $\mathbf{X}, \mathbf{Y}$ have identical moments (i.e. identic MGF) they must unavoidably have identical CDF.
- We can say more. If $\mathbf{Y} = a\mathbf{X} + b$, then $M_{\mathbf{Y}}(t) = \mathbb{E}(e^{(a\mathbf{X}+b)t}) = e^{bt} M_{\mathbf{X}}(t)$.
- Besides, if $\mathbf{Y}$ the sum of N independent, identically distributed random variables, $\mathbf{Y} = \mathbf{X}_1 + \mathbf{X}_2 + \dots$ then $M_{\mathbf{Y}}(t) = \mathbb{E}(e^{t(\mathbf{X}_1+\mathbf{X}_2+\dots)}) = \mathbb{E}(e^{t\mathbf{X}_1})\mathbb{E}(e^{t\mathbf{X}_2}) \dots$, due to the independency. And, finally $M_{\mathbf{Y}}(t) = [\mathbb{E}(e^{t\mathbf{X}})]^N = [M_{\mathbf{X}}(t)]^N$ since they are identically distributed.
- One last useful result related to MGF is that if $\mathbf{X} \sim \mathcal{N}(0, 1)$ then $M_{\mathbf{X}}(t) = e^{\frac{t^2}{2}}$, since the given distribution has mean = 0, variance = 1, skew = 0 and kurtosis = 1.

- Coming back to the proof of the CLT, we now understand that we need to prove that the MGF of $\bar{Z}_N$ converges to the MGF of $\mathcal{N}(0, 1)$, i.e.:

$$\lim_{N \rightarrow \infty} M_{\bar{Z}_N}(t) = e^{\frac{t^2}{2}}$$

- Let's define $Z_k = \frac{X_k - \mu}{\sigma}$. Then, $\bar{Z}_N = \frac{1}{\sqrt{N}} \sum_{k=1}^N Z_k$. We may write:

$$M_{\bar{Z}_N}(t) = M_{\frac{1}{\sqrt{N}} \sum_{k=1}^N Z_k}(t) = M_{\frac{Z_1}{\sqrt{N}} + \frac{Z_2}{\sqrt{N}} + \dots}(t)$$

- The various $Z_k/\sqrt{N}$ are, by construction, independent random variable, so that:

$$M_{\bar{Z}_N}(t) = [M_{Z/\sqrt{N}}(t)]^N = [M_Z(t/\sqrt{N})]^N$$

- Now, let's create a Taylor series expansion of $M_Z(t/\sqrt{N})$ at $t = 0$:

$$M_Z(t/\sqrt{N}) = \sum_{k=0}^{\infty} M_Z^k(0) \frac{(t/\sqrt{N})^k}{k!}$$

- Next, we split this expansion into two parts. The first part is a finite series of three terms corresponding to $k = 0$, $k = 1$, and $k = 2$. The second part is the remainder of the infinite series.

$$M^0$$

- , M^1 , M^2 , etc. are the 0-th, 1st, 2nd, and so on derivatives of the MGF $M_Z(t/\sqrt{N})$ evaluated at $(t/\sqrt{N}) = 0$. We know that these derivatives of the MGF are the 0-th, 1st, 2nd, etc. moments of Z .

- The 0-th moment, $M^0(0)$, is always 1. Z is, by its construction, a standard normal random variable. Hence, its first moment (mean), $M^1(0) = 0$, and its second moment (variance), $M^2(0) = 1$. So:

$$M_Z(t/\sqrt{N}) = 1 + 0 + \frac{(t/\sqrt{N})^2}{2!} + \sum_{k=3}^{\infty} M_Z^k(0) \frac{(t/\sqrt{N})^k}{k!}$$

- Now, let's summarize where we are:

$$\lim_{N \rightarrow \infty} M_{\bar{Z}_N}(t) = \lim_{N \rightarrow \infty} \left(M_Z(t/\sqrt{N}) \right)^N$$

$$\lim_{N \rightarrow \infty} M_{\bar{Z}_N}(t) = \lim_{N \rightarrow \infty} \left(1 + 0 + \frac{(t/\sqrt{N})^2}{2!} + R_2(t/\sqrt{N}) \right)^N$$

- Where $R_r(x)$ is the remainder or residual from approximating a function using the Taylor polynomial of order r .

- Taylor's theorem states that $\lim_{x \rightarrow a} \frac{R_r(x)}{(x-a)^r} = 0$. In our case this turns out to be:

$$\lim_{\frac{t}{\sqrt{N}} \rightarrow a} \frac{R_2(t/\sqrt{N})}{(t/\sqrt{N} - a)^2} = 0$$

- But, with $\alpha = 0$, this is equivalent to the limit:

$$\lim_{N \rightarrow \infty} \frac{R_2(t/\sqrt{N})}{(t/\sqrt{N})^2} = 0$$

- In the above limit, the L.H.S. will tend to zero as long as N tends to infinity independent of what value t has as long as it's finite.
- Therefore, we can write:

$$\lim_{N \rightarrow \infty} nR_2(t/\sqrt{N}) = 0$$

- With little algebra, we have that:

$$\lim_{N \rightarrow \infty} M_{\bar{Z}_N}(t) = \lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} [(t^2/2) + NR_2(t/\sqrt{N})] \right)^N$$

- And, therefore:

$$\lim_{N \rightarrow \infty} M_{\bar{Z}_N}(t) = \lim_{N \rightarrow \infty} \left(1 + \frac{(t^2/2)}{N} \right)^N$$

- Since $\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N} \right)^N = e^x$, we can finally write:

$$\lim_{N \rightarrow \infty} M_{\bar{Z}_N}(t) = e^{\frac{t^2}{2}}$$

- And the CLT is proved!

Credits

Course Flow

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